

SURBHI GOEL

[first name][last initial]@cis.upenn.edu

<https://www.surbhigoel.com>

Amy Gutmann Hall #623 ◊ 3333 Chestnut St ◊ Philadelphia, PA 19104

EDUCATION

The University of Texas at Austin August 2015 - June 2020

M.S. and Ph.D. in Computer Science

Advisor: Adam R. Klivans

Committee: Alex Dimakis, Raghu Meka, Eric Price

Dissertation: [Towards Provably Efficient Algorithms for Learning Neural Networks](#)

Received the Bert Kay Dissertation Award

Indian Institute of Technology, Delhi July 2011 - May 2015

B.Tech. in Computer Science and Engineering

Received the ICIM Stay Ahead and Suresh Chandra Memorial Trust Award

ACADEMIC POSITIONS HELD

University of Pennsylvania, Philadelphia, PA January 2023 - Present

Magerman Term Assistant Professor, Computer and Information Science

Simons Institute for Theory of Computing, Berkeley, CA August - December 2024

Program Co-organizer, Special Year on LLMs and Transformers

Visiting Scientist, Modern Paradigms of Generalization

Institute for Advanced Study, Princeton, NJ January - May 2020

Visiting Graduate Student, Theoretical Machine Learning Program

Simons Institute for Theory of Computing, Berkeley, CA May - August 2019

Research Fellow, Foundations of Deep Learning Program

University of Michigan, Ann Arbor, MI May - July 2013

Research Scholar

Supervisor: Atul Prakash

INDUSTRY POSITIONS HELD

Microsoft Research, New York, NY July 2020 - December 2022

Postdoctoral Researcher, Machine Learning Group

Google, Mountain View, CA May - August 2018

Research Intern

Supervisor: Rina Panigrahy

Dell, Round Rock, TX June - August 2017

Research Intern

Google, New York, NY May - August 2016

Research Intern

Supervisor: Natalia Ponomareva

Google, Mountain View, CA May - August 2014

Software Engineering Intern

Supervisor: Neha Jha

SELECTED AWARDS AND FELLOWSHIPS

CAREER Award <i>Faculty Early Career Development Program, National Science Foundation</i>	2026
Alfred P. Sloan Research Fellowship in Computer Science <i>Alfred P. Sloan Foundation</i>	2026
Best Paper Award <i>Workshop on AI for Mechanism Design and Strategic Decision Making, ICLR</i>	2026
AI2050 Early Career Fellowship <i>Schmidt Sciences</i>	2025
Best Paper Award <i>Reliable ML Workshop, NeurIPS</i>	2025
Amazon Research Award <i>Amazon AI</i>	2025
OpenAI Superalignment Fast Grant <i>OpenAI</i>	2024
Microsoft Accelerate Foundation Models Research Award <i>Microsoft Research</i>	2023
Bert Kay Dissertation Award <i>Best Dissertation, Department of Computer Science, UT Austin</i>	2020
Rising Star in Machine Learning <i>University of Maryland</i>	2019
Rising Star in EECS <i>UIUC</i>	2019
AI PhD Fellowship <i>J.P. Morgan</i>	2019
Simons-Berkeley Research Fellowship <i>Foundations of Deep Learning program, Simons Institute for Theory of Computing, UC Berkeley</i>	2019
ICIM Stay Ahead and Suresh Chandra Memorial Trust Award <i>Best Undergraduate Thesis, Indian Institute of Technology, Delhi</i>	2015
Aditya Birla Scholarship <i>Awarded to 12 students every year across India</i>	2011

RESEARCH GRANTS

A Full-Stack Science of AI Reliability <i>\$600,000, NSF CAREER Award</i>	June 2026 - May 2031
Alfred P. Sloan Research Fellowship <i>\$75,000, Alfred P. Sloan Foundation</i>	September 2026 - August 2028
Theoretical Foundations for Provably Safe Interactive AI <i>\$494,925, AI2050 Early Career Fellowship, Schmidt Sciences</i>	November 2025 - March 2029
Efficient and Safe Protocols for Collaborative Agentic AI <i>\$70,000 (+ \$50,000 compute credits), Amazon Research Award</i>	November 2025
Efficient Algorithms for Learning with Distribution Shift <i>\$700,000, NSF-Medium, with Adam R. Klivans</i>	August 2025 - July 2029

Principled Inference-Time Guardrails

December 2025 - May 2026

\$199,875, AISI Challenge Fund, with Eric Wong

Protocols for Safe Interaction with Multiple Misaligned AIs

September 2025 - May 2026

\$174,312, AISI Challenge Fund, with Aaron Roth

Mechanisms to Predict Emergent Behavior in LLMs

April 2024

\$150,000, OpenAI Superalignment Fast Grant, with Eran Malach

Improving OOD Performance of LLMs on Reasoning Tasks

September 2023

\$25,000 compute credits, Microsoft Accelerate Foundation Models Research Award

PUBLICATIONS

$(\alpha\text{-}\beta)$ indicates alphabetical ordering, as is traditional for theory papers.

WORKING PAPERS

- [P1] $(\alpha\text{-}\beta)$ Natalie Collina, Surbhi Goel, Aaron Roth, Sikata Bela Sengupta. **Delegating Authorization to Misaligned Agents: Coalitional Alignment and Safe Control.**
- [P2] $(\alpha\text{-}\beta)$ Surbhi Goel, Jonathan Pei, James Wang. **Learning When to Stop: Selective Imitation Learning Under Arbitrary Dynamics Shift.**
- [P3] Mirah Shi, Marcel Hussing, Natalie Collina, Ira Globus-Harris, Aaron Roth, Surbhi Goel. **Narrowing the Collaboration Gap, Probably.**
- [P4] $(\alpha\text{-}\beta)$ Natalie Collina, Surbhi Goel, Aaron Roth, Mirah Shi. **Personalization Aids Pluralistic Alignment Under Competition.** *Best paper award*, Workshop on AI for Mechanism Design and Strategic Decision Making, ICLR 2026.
- [P5] $(\alpha\text{-}\beta)$ Ezra Edelman, Surbhi Goel. **Reliable Abstention under Adversarial Injections: Tight Lower Bounds and New Upper Bounds.** FORC 2026 (Highlight track).
- [P6] Yufa Zhou, Yixiao Wang, Surbhi Goel, Anru Zhang. **Why Do Transformers Fail to Forecast Time Series In-Context?** *Spotlight presentation*, What Can('t) Transformers Do? Workshop, NeurIPS 2025.
- [P7] James Wang, Surbhi Goel. **Weight Clipping for Robust Conformal Inference under Unbounded Covariate Shifts.**

CONFERENCE PAPERS

- [C8] $(\alpha\text{-}\beta)$ Surbhi Goel, Adam R. Klivans, Konstantinos Stavropoulos, Arsen Vasilyan. **Testing Noise Assumptions of Learning Algorithms.** COLT 2026. *Best paper award*, Reliable ML Workshop, NeurIPS 2025.
- [C9] $(\alpha\text{-}\beta)$ Eric Eaton, Surbhi Goel, Marcel Hussing, Michael Kearns, Aaron Roth, Sikata Bela Sengupta, Jessica Sorrell. **Model Agreement via Anchoring.** COLT 2026.
- [C10] $(\alpha\text{-}\beta)$ Natalie Collina, Surbhi Goel, Aaron Roth, Emily Ryu, Mirah Shi. **Emergent Alignment via Competition.** ICML 2026.
- [C11] Jingwen Liu, Ezra Edelman, Surbhi Goel, Bingbin Liu. **Less Data, Faster Training: Repeating Smaller Datasets Speeds up Learning via Sampling Biases.** ICML 2026. *Contributed talk*, Sci4DL Workshop, ICLR 2026.
- [C12] Terry Tong, Yu Feng, Surbhi Goel, Dan Roth. **Is Code Better Than Language for Algorithmic Reasoning.** ICML 2026.

- [C13] Abhishek Panigrahi, Bingbin Liu, Sadhika Malladi, Sham Kakade, [Surbhi Goel](#). **In Good GRACEs: Principled Teacher Selection for Knowledge Distillation**. **ICLR 2026**.
- [C14] (α - β) Natalie Collina, Ira Globus-Harris, [Surbhi Goel](#), Varun Gupta, Aaron Roth, Mirah Shi. **Collaborative Prediction: Tractable Information Aggregation via Agreement**. **SODA 2026**. *Spotlight presentation, EC 2025 Workshop on Human-AI Collaboration. Spotlight presentation, 2025 TTIC Workshop on Incentives for Collaborative Learning and Data Sharing*.
- [C15] Helen Jin, Anton Xue, Weiqiu You, [Surbhi Goel](#), Eric Wong. **Probabilistic Stability Guarantees for Feature Attributions**. **NeurIPS 2025**.
- [C16] Nirmal Joshi, Gal Vardi, Adam Block, [Surbhi Goel](#), Zhiyuan Li, Theodor Misiakiewicz, Nati Srebro. **A Theory of Learning with Autoregressive Chain of Thought**. **COLT 2025**.
- [C17] (α - β) Natalie Collina, [Surbhi Goel](#), Varun Gupta, Aaron Roth. **Tractable Agreement Protocols**. **STOC 2025**.
- [C18] Maxon Rubin-Toles, Maya Gambhir, Keshav Ramji, Aaron Roth, [Surbhi Goel](#). **Conformal Language Model Reasoning with Coherent Factuality**. **ICLR 2025**.
- [C19] Abhishek Panigrahi, Bingbin Liu, Sadhika Malladi, Andrej Risteski, [Surbhi Goel](#). **Progressive Distillation Induces an Implicit Curriculum**. *Oral presentation*, **ICLR 2025**.
- [C20] Anton Xue, Avishree Khare, Rajeev Alur, [Surbhi Goel](#), Eric Wong. **Logicbreaks: A Framework for Understanding Subversion of Rule-based Inference**. **ICLR 2025**.
- [C21] Ezra Edelman, Nikolaos Tsilivis, Benjamin L. Edelman, Eran Malach, [Surbhi Goel](#). **The Evolution of Statistical Induction Heads: In-Context Learning Markov Chains**. **NeurIPS 2024**.
- [C22] (α - β) [Surbhi Goel](#), Abhishek Shetty, Konstantinos Stavropoulos, Arsen Vasilyan. **Tolerant Algorithms for Learning with Arbitrary Covariate Shift**. *Spotlight presentation*, **NeurIPS 2024**.
- [C23] GuanWen Qiu, Da Kuang, [Surbhi Goel](#). **Complexity Matters: Feature Learning in the Presence of Spurious Correlations**. **ICML 2024**.
- [C24] Kan Xu, Hamsa Bastani, [Surbhi Goel](#), Osbert Bastani. **Stochastic Bandits with ReLU Neural Networks**. **ICML 2024**.
- [C25] (α - β) [Surbhi Goel](#), Steve Hanneke, Shay Moran, Abhishek Shetty. **Adversarial Resilience in Sequential Prediction via Abstention**. **NeurIPS 2023**.
- [C26] (α - β) Benjamin L. Edelman, [Surbhi Goel](#), Sham M. Kakade, Eran Malach, Cyril Zhang. **Pareto Frontiers in Neural Feature Learning: Data, Compute, Width, and Luck**. *Spotlight presentation*, **NeurIPS 2023**.
- [C27] Bingbin Liu, Jordan T. Ash, [Surbhi Goel](#), Akshay Krishnamurthy, Cyril Zhang. **Exposing Attention Glitches with Flip-Flop Language Modeling**. *Spotlight presentation*, **NeurIPS 2023**.
- [C28] (α - β) Sitan Chen, Zehao Dou, [Surbhi Goel](#), Adam R. Klivans, Raghu Meka. **Learning Narrow One-Hidden-Layer ReLU Networks**. **COLT 2023**.
- [C29] Bingbin Liu, Jordan T. Ash, [Surbhi Goel](#), Akshay Krishnamurthy, Cyril Zhang. **Transformers Learn Shortcuts to Automata**. *Notable top-5% paper*, **ICLR 2023**.
- [C30] (α - β) [Surbhi Goel](#), Sham M. Kakade, Adam T. Kalai, Cyril Zhang. **Recurrent Convolutional Neural Networks Learn Succinct Learning Algorithms**. **NeurIPS 2022**.

- [C31] (α - β) Boaz Barak, Benjamin L. Edelman, Surbhi Goel, Sham M. Kakade, Eran Malach, Cyril Zhang. **Hidden Progress in Deep Learning: SGD Learns Parities Near the Computational Limit.** *NeurIPS 2022*.
- [C32] (α - β) Benjamin L. Edelman, Surbhi Goel, Sham M. Kakade, Cyril Zhang. **Inductive Biases and Variable Creation in Self-Attention Mechanisms.** *ICML 2022*.
- [C33] Nikunj Saunshi, Jordan T. Ash, Surbhi Goel, Dipendra Misra, Cyril Zhang, Sanjeev Arora, Sham M. Kakade, Akshay Krishnamurthy. **Understanding Contrastive Learning Requires Incorporating Inductive Biases.** *ICML 2022*.
- [C34] Jordan T. Ash, Cyril Zhang, Surbhi Goel, Akshay Krishnamurthy, Sham M. Kakade. **Anti-Concentrated Confidence Bonuses For Scalable Exploration.** *ICLR 2022*.
- [C35] (α - β) Jordan T. Ash, Surbhi Goel, Akshay Krishnamurthy, Dipendra Misra. **Investigating the Role of Negatives in Contrastive Representation Learning.** *AISTATS 2022*.
- [C36] Jordan T. Ash, Surbhi Goel, Akshay Krishnamurthy, Sham M. Kakade. **Gone Fishing: Neural Active Learning with Fisher Embeddings.** *NeurIPS 2021*.
- [C37] (α - β) Naman Agarwal, Surbhi Goel, Cyril Zhang. **Acceleration via Fractal Learning Rate Schedules.** *ICML 2021*.
- [C38] (α - β) Yuval Dagan, Constantinos Daskalakis, Nishanth Dikkala, Surbhi Goel, Vardis Kandiros. **Statistical Estimation from Dependent Data.** *ICML 2021*.
- [C39] (α - β) Surbhi Goel, Adam R. Klivans, Pasin Manurangsi, Daniel Reichman. **Tight Hardness Results for Training Depth-2 ReLU Networks.** *ITCS 2021*.
- [C40] (α - β) Surbhi Goel, Adam R. Klivans, Frederic Koehler. **From Boltzmann Machines to Neural Networks and Back Again.** *NeurIPS 2020*.
- [C41] (α - β) Surbhi Goel, Aravind Gollakota, Adam R. Klivans. **Statistical-Query Lower Bounds via Functional Gradients.** *NeurIPS 2020*.
- [C42] (α - β) Surbhi Goel, Aravind Gollakota, Zhihan Jin, Sushrut Karmalkar, Adam R. Klivans. **Super-polynomial Lower Bounds for Learning One-Layer Neural Networks using Gradient Descent.** *ICML 2020*.
- [C43] Omar Montasser, Surbhi Goel, Ilias Diakonikolas, Nathan Srebro. **Efficiently Learning Adversarially Robust Halfspaces with Noise.** *ICML 2020*.
- [C44] Jessica Hoffmann, Soumya Basu, Surbhi Goel, Constantine Caramanis. **Learning Mixtures of Graphs from Epidemic Cascades.** *ICML 2020*.
- [C45] (α - β) Ilias Diakonikolas, Surbhi Goel, Sushrut Karmalkar, Adam R. Klivans, Mahdi Soltanolkotabi. **Approximation Schemes for ReLU Regression.** *COLT 2020*.
- [C46] Surbhi Goel. **Learning Ising and Potts Models with Latent Variables.** *AISTATS 2020*.
- [C47] (α - β) Surbhi Goel, Sushrut Karmalkar, Adam R. Klivans. **Time/Accuracy Trade-offs for Learning a ReLU with respect to Gaussian Marginals.** *Spotlight presentation, NeurIPS 2019*.
- [C48] (α - β) Surbhi Goel, Daniel Kane, Adam R. Klivans. **Learning Ising Models with Independent Failures.** *COLT 2019*.
- [C49] (α - β) Surbhi Goel, Adam R. Klivans. **Learning Neural Networks with Two Nonlinear Layers in Polynomial Time.** *COLT 2019. Deep Learning: Bridging Theory and Practice Workshop, NeurIPS 2017*.

- [C50] (α - β) Surbhi Goel, Adam R. Klivans, Raghu Meka. **Learning One Convolutional Layer with Overlapping Patches.** *Oral presentation*, ICML 2018.
- [C51] (α - β) Surbhi Goel, Adam R. Klivans. **Eigenvalue Decay Implies Polynomial-Time Learnability for Neural Networks.** NeurIPS 2017.
- [C52] (α - β) Surbhi Goel, Varun Kanade, Adam R. Klivans, Justin Thaler. **Reliably Learning the ReLU in Polynomial Time.** COLT 2017. *Oral presentation*, *Optimization for Machine Learning (OPT-ML) Workshop, NeurIPS 2016*.

MANUSCRIPTS

- [M53] Mahdi Sabbaghi, George J. Pappas, Hamed Hassani, Surbhi Goel. **Explicitly Encoding Structural Symmetry is Key to Length Generalization in Arithmetic Tasks.** Manuscript, 2024.
- [M54] (α - β) Surbhi Goel, Rina Panigrahy. **Recovering the Lowest Layer of Deep Networks with High Threshold Activations.** Manuscript, 2019.
- [M55] Matthew Jordan, Naren Manoj, Surbhi Goel, Alexandros Dimakis. **Quantifying Perceptual Distortion of Adversarial Examples.** Manuscript, 2019.
- [M56] (α - β) Simon Du, Surbhi Goel. **Improved Learning of One-hidden-layer Convolutional Neural Networks with Overlaps.** Manuscript, 2018.

SELECTED MEDIA COVERAGE AND INTERVIEWS

- **How UT Computer Science Shaped a Sloan Fellow: Q&A with Surbhi Goel.**
UT Austin Computer Science, June 5, 2026.
Interview on trustworthy AI research, the Sloan Research Fellowship, and mentorship.
- **We are losing the point of education.**
The Daily Pennsylvanian, February 8, 2026.
Guest column by Amogh Sarangdhar highlighting oral assessments in CIS 5200 (Machine Learning) as a response to AI use in education.
- **Penn professor named 2025 Schmidt Sciences AI2050 Early Career Fellow.**
The Daily Pennsylvanian, November 7, 2025.
Profile of the fellowship and research on the theoretical foundations of reliable AI.
- **Surbhi Goel Explains How ChatGPT Actually Works.**
Penn Engineering, April 23, 2024.
Feature on large language models and the Women in Data Science @ Penn presentation.

INVITED TALKS

- [T1] *Reliable Learning via Abstention*. Mathematical Foundations of Artificial Intelligence (MFAI), UNAM, Mexico. April 2026.
- [T2] *Reliable Learning via Design: From Abstention to Collaboration*. HDSI, UCSD. April 2026.
- [T3] *Reliable Learning via Abstention*. Statistics Seminar, MIT. April 2026.
- [T4] *Reliable Learning via Abstention*. Statistics Seminar, Yale. April 2026.
- [T5] *Reliable Learning via Abstention*. Alg-ML Seminar, Princeton. March 2026.
- [T6] *Reliable Learning via Abstention*. Plenary Talk, ALT. February 2026.
- [T7] *Human-AI Collaboration via Agreement and Competition*. Modern Paradigms of Generalization Reunion, Simons Institute. January 2026.

- [T8] *Reliable Human-AI Collaboration via Agreement and Competition*. Reliable ML Workshop, NeurIPS. December 2025.
- [T9] *Collaborative Prediction via Tractable Agreement Protocols*. DeepMath, University of Michigan in Ann Arbor. November 2025.
- [T10] *Collaborative Prediction via Tractable Agreement Protocols*. Data Science: Probabilistic and Optimization Methods II, ICTS, India. August 2025.
- [T11] *Collaborative Prediction via Tractable Agreement Protocols*. Harvard Theory Seminar. April 2025.
- [T12] *Synthetic Tasks as Sandboxes for Attributing Model Behavior*. TILOS-HDSI Seminar, UCSD. March 2025.
- [T13] *Synthetic Tasks as Sandboxes for Attributing Model Behavior*. ATTRIB Workshop, NeurIPS. December 2024.
- [T14] *Tractable Agreement Protocols*. Stanford Theory Seminar. November 2024.
- [T15] *Algorithms for Learning under Arbitrary Covariate Shift*. TOCA-SV, Stanford. November 2024.
- [T16] *Beyond Worst-case Sequential Prediction: Adversarial Robustness via Abstention*. Emerging Paradigms of Generalization Workshop, Simons Institute. September 2024.
- [T17] *How do Large Language Models Think?*. Penn Engineering’s AI for Executives: Daylong Immersion. May 2024.
- [T18] *Beyond Worst-case Sequential Prediction: Adversarial Robustness via Abstention*. Rutgers/DIMACS Theory Seminar. April 2024.
- [T19] *Beyond Worst-case Sequential Prediction: Adversarial Robustness via Abstention*. MINDS Seminar, JHU. March 2024.
- [T20] *Understanding Training Dynamics in Deep Learning using Simplified Models*. Optimization Seminar, UPenn. March 2024.
- [T21] *Beyond Worst-case Sequential Prediction: Adversarial Robustness via Abstention*. EnCORE Workshop, IPAM, UCLA. March 2024.
- [T22] *How do Large Language Models Think?*. Women in Data Science, UPenn. February 2024.
- [T23] *Thinking fast with Transformers: Algorithmic Reasoning via Shortcuts*. Deep Learning Down Under Workshop, Lorne, Australia. January 2024.
- [T24] *Beyond Worst-case Sequential Prediction: Adversarial Robustness via Abstention*. Theory Seminar, UPenn. November 2023.
- [T25] *Thinking fast with Transformers: Algorithmic Reasoning via Shortcuts*. IFML Workshop on Generative AI, UT Austin. November 2023.
- [T26] *Beyond Worst-case Sequential Prediction: Adversarial Robustness via Abstention*. Alg-ML Seminar, Princeton. November 2023.
- [T27] *Beyond Worst-case Sequential Prediction: Adversarial Robustness via Abstention*. BLISS Seminar, UC Berkeley. October 2023.
- [T28] *Beyond Worst-case Sequential Prediction: Adversarial Robustness via Abstention*. Math Machine Learning Seminar, MPI MIS + UCLA. August 2023.
- [T29] *Beyond Worst-case Sequential Prediction: Adversarial Robustness via Abstention*. FODSI Workshop on Computational Complexity of Statistical Problems, MIT. June 2023.

- [T30] *Thinking fast with Transformers: Algorithmic Reasoning via Shortcuts*. Youth in High Dimensions, Trieste, Italy. May 2023.
- [T31] *Thinking fast with Transformers: Algorithmic Reasoning via Shortcuts*. MaD Seminar, NYU. April 2023.
- [T32] *Thinking fast with Transformers: Algorithmic Reasoning via Shortcuts*. ASSET Seminar, UPenn. April 2023.
- [T33] *What Functions do Self-attention Blocks Prefer to Represent? Demystifying Attention-based Architectures in Deep Learning*. Joint IFML/Data-Driven Decision Processes Workshop, Simons Institute. October 2022.
- [T34] *Sparse Feature Emergence in Deep Learning*. Symposium on New Directions in Theoretical Machine Learning. September 2022.
- [T35] *What Functions do Self-attention Blocks Prefer to Represent? Demystifying Attention-based Architectures in Deep Learning*. ML Foundations Seminar at MSR Redmond. August 2022.
- [T36] *The Hidden Progress Behind Loss Curves*. Workshop on Learning: Optimization and Stochastics at EPFL. July 2022.
- [T37] *What Functions do Self-attention Blocks Prefer to Represent? Demystifying Attention-based Architectures in Deep Learning*. Workshop on Algorithms for Learning and Economics (WALE) in Greece. June 2022.
- [T38] *Principled Algorithm Design in the Era of Deep Learning*. CS/CSE Colloquium at NYU Courant/Tandon. April 2022.
- [T39] *Principled Algorithm Design in the Era of Deep Learning*. CS Colloquium, UW-Madison. March 2022.
- [T40] *Principled Algorithm Design in the Era of Deep Learning*. CS Colloquium, HDSI, UCSD. March 2022.
- [T41] *Principled Algorithm Design in the Era of Deep Learning*. CS Colloquium, UMD. February 2022.
- [T42] *Principled Algorithm Design in the Era of Deep Learning*. SCS Talk, CMU. February 2022.
- [T43] *Principled Algorithm Design in the Era of Deep Learning*. CS Colloquium, Duke. February 2022.
- [T44] *Principled Algorithm Design in the Era of Deep Learning*. CIS Colloquium, UPenn. February 2022.
- [T45] *Principled Algorithm Design in the Era of Deep Learning*. CS Colloquium, Cornell. February 2022.
- [T46] *Principled Algorithm Design in the Era of Deep Learning*. Talks at TTIC. February 2022.
- [T47] *What Functions do Self-attention Blocks Prefer to Represent? Demystifying Attention-based Architectures in Deep Learning*. ML Symposium, USC. December 2021.
- [T48] *What Functions do Self-attention Blocks Prefer to Represent? Demystifying Attention-based Architectures in Deep Learning*. ELLIS Talk Series, IST Austria. December 2021.
- [T49] *What Functions do Self-attention Blocks Prefer to Represent? Demystifying Attention-based Architectures in Deep Learning*. Learning Theory Workshop, Google. October 2021.
- [T50] *Computational Barriers For Learning Some Generalized Linear Models*. Information-Computation Trade-offs Workshop, Simons Institute. September 2021.

- [T51] *Computational Complexity of Learning Neural Networks over Gaussian Marginals*. Statistics Seminar, Stanford University. July 2021.
- [T52] *Computational Complexity of ReLU Regression*. The Multifaceted Complexity of Machine Learning Workshop, IMSI. April 2021.
- [T53] *Computational Complexity of Learning Neural Networks over Gaussian Marginals*. SILO Seminar, UW-Madison. January 2021.
- [T54] *Computational Complexity of Learning Neural Networks over Gaussian Marginals*. TOC Colloquium, MIT. December 2020.
- [T55] *Computational Complexity of Learning Neural Networks over Gaussian Marginals*. IDEAL Seminar, TTIC. November 2020.
- [T56] *Computational Complexity of Learning Neural Networks over Gaussian Marginals*. ARC Colloquium, Georgia Tech. November 2020.
- [T57] *Computational Complexity of Learning Neural Networks over Gaussian Marginals*. ML Theory Seminar, Harvard University. October 2020.
- [T58] *Computational Complexity of Learning Neural Networks over Gaussian Marginals*. Algorithms Seminar, Duke University. October 2020.
- [T59] *Approximation Schemes for ReLU Regression*. Deep Learning Program Reunion at Simons Institute. August 2020.
- [T60] *Computational Complexity of Learning Neural Networks over Gaussian Marginals*. MIC Seminar at NYU. May 2020.
- [T61] *Provably Efficient Algorithms for Learning Neural Networks*. Microsoft Research New York. February 2020.
- [T62] *Provably Efficient Algorithms for Learning Neural Networks*. Microsoft Research New England. February 2020.
- [T63] *Provably Efficient Algorithms for Learning Neural Networks*. Microsoft Research Redmond. February 2020.
- [T64] *Exploring Surrogate Losses for Learning Neural Networks*. TTIC Young Researcher Seminar Series. December 2019.
- [T65] *Efficiently Learning Simple Neural Networks*. Rising Star in ML Talk at University of Maryland. September 2019.
- [T66] *Learning Ising Models with Independent Failures*. Research Fellows Talk at Simons Institute. July 2019.
- [T67] *Efficiently Learning Simple Convolutional Networks*. China Theory Week at Tsinghua University. September 2018.
- [T68] *Learning One Convolutional Layer with Overlapping Patches*. Google Research Theory Reading Group. June 2018.

TEACHING

CIS 6200: Advanced Topics in Machine Learning

Fall 2026

Emerging Theory of Large Language Models

Instructor

University of Pennsylvania

CIS 5200: Machine Learning*Co-instructor with Eric Wong*

Spring 2023, 2024, 2025, 2026

*University of Pennsylvania***CIS 3333: Mathematics of Machine Learning***Instructor*

Fall 2025

*University of Pennsylvania***CIS 7000: Foundations of Modern ML - Theory and Empirics***Instructor*

Fall 2023

*University of Pennsylvania***DISSERTATIONS/THESES SUPERVISED**

Sophia Pi (PhD)

Fall 2026 - Present

*Co-advisors: Meena Jagadeesan, Michael Kearns, and Aaron Roth**Thesis: Anticipated Spring 2031***Berkan Ottlik (PhD)**

Fall 2025 - Present

*Thesis: Anticipated Spring 2030**NSF Graduate Research Fellowship***Honam Wong (PhD)**

Fall 2025 - Present

*Co-advisor: Enric Boix-Adserà**Thesis: Anticipated Spring 2030**AWS ASSET Fellow, 2026***Ezra Edelman (PhD)**

Fall 2023 - Present

*Thesis: Anticipated Spring 2028**AWS ASSET Fellow, 2025***Aaron Mei (Undergraduate)**

Spring 2025

*Thesis: Mechanistically Understanding the Internal Computations of Chess LLMs***Dante Lokitayakul (Master's)**

Fall 2023 - Fall 2024

*Thesis: Architectural Inductive Biases and Moment Alignment: Analyzing Representational Capacity in Self-Supervised Learning**Next Position: Machine Learning Engineer at Prompt Inversion***Nimay Kumar (Master's)**

Fall 2023

*Thesis: Automatic Data Augmentation Methods for Contrastive Self-Supervised Learning**Next Position: Data Scientist at TD Bank***DOCTORAL COMMITTEE MEMBERSHIPS**

Kyurae Kim

Expected Spring 2027

*Advisor: Jake Gardner***Yiqiao Bao**

Expected Spring 2027

*PhD student, Computer and Information Science, University of Pennsylvania**Advisors: Anindya De and Sampath Kannan***Salma Tarmoun**

Expected Fall 2026

Advisor: René Vidal

Natalie Collina	2026
<i>Advisors: Michael Kearns and Aaron Roth</i>	
<i>Title: Learning to Play Well with Others</i>	
<i>Next Position: Postdoc at MIT</i>	
Behrad Moniri	2026
<i>Advisor: Hamed Hassani</i>	
<i>Title: Understanding Deep Learning via Tractable Models</i>	
<i>Next Position: Machine Learning Researcher at Cubist Systematic Strategies (Point72)</i>	
Weiqliu You	2026
<i>Advisor: Eric Wong</i>	
<i>Title: Structured Explanations for Scientific Domains Building on Foundation Models</i>	
Rubing Yang	2026
<i>Advisor: Pratik Chaudhari</i>	
<i>Title: The Impact of Data on Generalization in Deep Learning</i>	
Abhishek Panigrahi (Princeton University)	May 2026
<i>Dissertation reader; Advisor: Sanjeev Arora</i>	
<i>Title: Understanding Optimization, Distillation, and Transfer: A Theoretical Study of Deep Learning</i>	
<i>Next Position: Research Scientist at Google DeepMind</i>	
Tyler LaBonte (Georgia Tech)	Spring 2026
<i>Advisors: Vidya Muthukumar and Jake Abernathy</i>	
<i>Title: Generalization in Dynamic Environments: Foundations and Algorithms</i>	
<i>Next Position: Research Scientist at Microsoft Research</i>	
Anton Xue	Summer 2025
<i>Advisors: Rajeev Alur and Eric Wong</i>	
<i>Title: Trustworthy Machine Learning: Specification, Verification, and Explanation</i>	
<i>Next Position: Postdoc at UT Austin Center for Generative AI</i>	
Varun Gupta	Spring 2025
<i>Advisors: Michael Kearns and Aaron Roth</i>	
<i>Title: Calibrated Learning and Reliable Decision-Making</i>	
<i>Next Position: Postdoc at Vector Institute</i>	
Mingyuan Zhang	Spring 2024
<i>Advisor: Shivani Agarwal</i>	
<i>Title: Statistical Machine Learning for Complex Classification Problems</i>	
<i>Next Position: Senior AI Researcher at LinkedIn</i>	

STUDENTS SUPERVISED

David Zhang (Undergraduate at Penn)	
<i>Projects: Gradient-Based Data Selection for Instruction Tuning; In-Context Learning in Transformers</i>	
Megan Mann (Undergraduate)	Summer 2026
<i>Research Experience for Undergraduates (REU)</i>	

James Wang (Undergraduate at Penn) Fall 2025 - Present
Project: Conformal Prediction under Distribution Shifts
Next Position: PhD at Columbia

Jonathan Pei (Undergraduate at Penn) Spring 2025, Fall 2025 - Present
Projects: Algorithmic Generalization in LLMs; Reliable Learning under Distribution Shifts
Summer REU, 2026

Maya Gambhir (Undergraduate at Penn) Summer 2024; 2025 - 2026
Senior Thesis: Generating Synthetic Data to Evaluate Sortition Algorithms
Summer REU: Conformal Language Model Reasoning with Coherent Factuality
Next Position: PhD at Princeton
Received 2026 Wolf-Hallac Award

Abhishek Panigrahi (PhD Student at Princeton) Fall 2023 - Fall 2025
Project: Understanding the Benefits of Distillation from an Optimization Perspective

Jingwen Liu (PhD Student at Columbia) Summer 2025
Project: Learning Single-Index Models over Axis-Aligned Marginals

Yufa Zhou (Master's at Penn) Spring 2025
Independent Study: Limitations of Auto-regressive Models to Learn Time-Series Prediction Tasks
Next Position: PhD at Duke University

Niki Hasrati (Visiting Student)
Current Position: PhD student at CMU

Terry Tong (Undergraduate at UC Davis)
Project: Is Code Better than Language for Algorithmic Reasoning?
Next Position: PhD at Columbia, supported by NSF GRFP

Keshav Ramji (Undergraduate at Penn)
Project: Conformal Language Model Reasoning with Coherent Factuality
Next Position: Researcher at IBM Research

GuanWen Qiu (Master's at Penn) Spring 2023 - Spring 2024
Independent Study: Feature Learning Dynamics in the Presence of Spurious Correlations
Next Position: PhD at Purdue University

Alyssa Nie (Undergraduate at Penn) Fall 2023
Project: Towards a Mechanistic Understanding of Transformers on Flip-Flop Language Modeling

Maxon Rubin-Toles (Undergraduate at Penn) Summer 2024
Summer Research Internship: Conformal Language Model Reasoning with Coherent Factuality
Next Position: PhD at Harvard University, supported by NSF GRFP
Received 2025 Computer Science Academic Award

Bingbin Liu (PhD Student at CMU) Summer 2022
Internship at MSR: Quantifying how Transformers Learn Shortcuts to Reason
Next Position: Research Fellow at Simons Institute

Benjamin L. Edelman (PhD Student at Harvard) <i>Internship at MSR: Inductive Biases of Transformers</i> <i>Next Position: TechCongress AI Safety Fellow at the US AI Safety Institute</i>	Summer 2021
Nikunj Saunshi (PhD Student at Princeton) <i>Internship at MSR: Role of Architecture in the Success of Self-Supervised Learning</i> <i>Next Position: Research Scientist at Google</i>	Summer 2021

WORKSHOP AND PROGRAM ORGANIZATION

Workshop Co-organizer <i>Simons Institute's Multi-Program AI Reunion Workshop</i>	2026
Workshop Co-organizer <i>Methods and Opportunities at Small Scale (MOSS) at ICML</i>	2025
Program Co-organizer <i>Simons Institute's Special Year on LLMs and Transformers</i>	2023 - 2025
Workshop Co-organizer <i>Transformers as a Computational Model, Special Year on LLMs and Transformers, Simons Institute</i>	2024
Workshop Co-organizer <i>Unknown Futures of Generalization, Modern Paradigms in Generalization Program, Simons Institute</i>	2023 - 2024
Workshop Co-organizer <i>Bootcamp on LLMs and Transformers, Special Year on LLMs and Transformers, Simons Institute</i>	2024
Workshop Co-organizer <i>Mathematics of Modern Machine Learning (M3L) at NeurIPS 2023</i>	2023
Seminar Co-organizer <i>One World Machine Learning Seminar Series</i>	2020 - 2021

RESEARCH COMMUNITY SERVICE

Program Co-chair <i>Conference on Algorithmic Learning Theory (ALT 2027)</i>	2026 - Present
Workshop Co-chair <i>Conference on Language Modeling (COLM)</i>	2026
Steering Committee Member <i>Association for Algorithmic Learning Theory (AALT)</i>	2024 - Present
Co-treasurer <i>Association for Computational Learning (ACL)</i>	2024 - Present
Action Editor <i>Transactions on Machine Learning Research (TMLR)</i>	2024 - Present
Workshop Reviewing Committee <i>International Conference on Machine Learning (ICML)</i>	2024 - 2025
Office Hours Co-chair <i>International Conference on Learning Representations (ICLR)</i>	2023 - 2024
Virtual Experience Co-chair <i>Conference on Learning Theory (COLT)</i>	2023

Online Experience Co-chair	2021
<i>Conference on Learning Theory (COLT)</i>	
Program Committee	
<i>ICML (area chair)</i>	2026
<i>ALT (senior program committee)</i>	2024/25/26
<i>COLT (senior program committee)</i>	2024/25/26
<i>NeurIPS (area chair)</i>	2023/24/25/26
<i>AISTATS (area chair)</i>	2023
<i>ALT</i>	2021/22/23
<i>COLT</i>	2021/22
Conference Reviewing	
<i>STOC</i>	2019/20/21/25/26
<i>COLT</i>	2018/19/20
<i>ICLR</i>	2019/20/23
<i>SODA</i>	2020/23
<i>FOCS</i>	2020/22
<i>NeurIPS</i>	2018 (top 30%)/20/21
<i>ICML</i>	2019 (top 5%)
Journal Reviewing	
<i>TheoretiCS</i>	2024
<i>Journal of Machine Learning Research</i>	2021/22
<i>IEEE Transactions on Information Theory</i>	2020
Treasurer	2016 - 2017
<i>Graduate Representative Association of Computer Sciences (GRACS), UT Austin</i>	

PENN COMMUNITY SERVICE

Member	Fall 2025 - Spring 2026
<i>CIS Hiring Committee</i>	
Member	Fall 2025 - Present
<i>Faculty Committee on Undergraduate Research and Fellowships</i>	
Co-organizer	Summer 2025 - Present
<i>ASSET Seminar</i>	
Member	Fall 2024 - Spring 2025
<i>IDEAS Hiring Committee</i>	
Faculty Support	Fall 2024 - Present
<i>WiML@PennCIS</i>	
Speaker	May 2024
<i>AI for Executives: Daylong Immersion at Penn Engineering</i>	
Panelist	April 2024
<i>Innovation & Impact: Exploring AI in Engineering</i>	
Panelist	April 2024
<i>Special Webinar for AI Month</i>	
Speaker	February 2024
<i>Women in Data Science, Wharton</i>	

Member <i>CIS Hiring Committee</i>	Fall 2023 - Spring 2024
Co-organizer <i>Lunches with Women PhD Students at CIS</i>	Spring 2023
Co-organizer <i>Junior Faculty - PhD Lunches</i>	Spring 2023

MENTORING AND OUTREACH

Co-founder and Executive Committee Member <i>Learning Theory Alliance (LeT-All)</i>	2020 - Present
---	----------------

Co-organized a [social at NeurIPS 2025](#)
Co-organized the [Fall 2025 Mentoring Workshop](#)
Co-organized a [community event at COLT 2025](#)
Co-organized a [social at NeurIPS 2024](#)
Co-organized the [Fall 2023 Mentoring Workshop](#)
Co-organized the [Fall 2022 Mentoring Workshop](#) in collaboration with FODSI
Co-organized the [COLT 2022 Mentoring Panel](#)
Co-organized the [ALT 2022 Mentoring Workshop](#)
Co-organized the [Graduate Applications Support Program](#) in collaboration with WiML-T
Co-organized the [COLT 2021 Mentoring Workshop](#)
Co-organized the [ALT 2021 Mentoring Workshop](#)

Mentor <i>Women in Machine Learning Theory (WiML-T) Mentoring Program</i>	2021 - 2025
<i>UT Austin's Women in CS (GWC-WiCS) Mentoring Program</i>	2018 - 2019

Panelist <i>New in ML Workshop, NeurIPS 2023</i>	December 2023
<i>WiML Un-Workshop, ICML 2022</i>	July 2022
<i>New Horizons in Theoretical Computer Science</i>	June 2022
<i>VMware Nirman for Women in Tech</i>	January 2021